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3rd year 2nd Semester B.Sc. (Engg.) Final Examination, 2023

Course Title: Computer Networks   Course Code: ICT-3207

Time: 3 hours   Full Marks: 70

[Answer Any Five Questions]

1.	(a)	Define data and telecommunication. Write down the main components of a data communication.	4
	(b)	Consider a MESH topology network having 7 communication nodes are connected. Now determine the total minimum number of connected links that satisfy the MESH Topology concept for i) Half-duplex mode ii) Full Duplex Mode.	4
	(c)	Draw the ISO-OSI 7(seven) layers with their communication unit and briefly explain about Data link, Network, and Transport Layer	6
2.	(a)	TCP/IP reference model is more organized model compared to ISO-OSI model- do you agree? Explain with your answer by defining proper distinction.	4
	(b)	Write at least 3 (three) protocol names for network, transport and presentation layer	2
	(c)	Define Baseband and Broadband transmission. Write down the key distinctions between Baseband and Broadband transmission.	4
	(d)	A nonperiodic composite signal has a bandwidth of 200kHz, with a middle frequency of 140 kHz and peak amplitude of 20 V The two extreme frequencies have an amplitude of 0. Now Draw the frequency domain of the signal	4
3.	(a)	What is Multiplexing? Assume that a voice channel occupies a bandwidth of 5 kHz. We need to combine three voice channels into a link with a bandwidth of 17 kHz, from 20 to 37 kHz. Show the configuration, using the frequency domain. Assume there is a guard band of 1 KHz.	5
	(b)	Synchronous TDMA and statistical TDMA, which one is better, and Why? Explain your answer.	3
	(c)	A multiplexer combines four 100-kbps channels using a time slot of 2 bits. Show the output with four arbitrary inputs. What is the frame rate? What is the frame duration? What is the bit rate? What is the bit duration?	4
	(d)	Draw the delay analysis of a virtual circuit network.	2
4.	(a)	Draw and explain the different propagation modes used in Fiber Optic Communication.	4
	(b)	What do you know about virtual circuit network? Briefly explain about source-to-destination data transfer in virtual circuit network with appropriate illustration.	4
	(c)	Briefly describe the Hamming code encoder, decoder and logic analyzer with appropriate diagram.	6
5.	(a)	Define single-bit error and burst error. Determine the dataword-codeword C(7,4) table using the cyclic redundancy check(CRC) in aspects of CRC encoder operation for the following dataword. Assume the divisor is 1011. I. 0101 II. 0111 III. 1000 IV. 1010	5
	(b)	What is checksum? assume that, the sender wants to send five 4-bit numbers (7, 11, 12, 0, 6) to the receiver using checksum mechanism. Now explain the sender site and receiver site operation with appropriate diagram.	5
	(c)	Briefly explain the Selective Repeat ARQ protocol with appropriate illustration.	4

6. (a) Briefly explain the CSMA/CA protocol with flow and timing diagram. 4

(b) Why CSMS/CD cannot used in wireless LAN communication? A pure ALOHA network transmits 400-bit frames on a shared channel of 200 kbps What is the throughput if the system (all stations together) produces- 6

I. 2000 frames per second II. 1500 frames per second

III. 1000 frames per second IV. 500 frames per second.

(c) Show that in Go-Back-N ARQ, the size of the send window must be less than 2^m, where m is the size of the sequence number field in bits. 4

7. (a) Prove that in CDMA communication, a receiving station can get the data sent by a specific sender if it multiplies the entire data on the channel by the sender's chip code and then divides it by the number of stations 4

(b) Write down the key distinctions between IPv4 and IPv6. Find the class of each address. 5

I. 00000001.00001011.00001011.11101111

II. 11000001.10000011.00011011.11111111

III. 14.23.120.8

IV. 252.5.15.111

(c) An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows: 5

I. The first group has 64 customers; each needs 256 addresses.

II. The second group has 128 customers; each needs 128 addresses.

III. The third group has 128 customers; each needs 64 addresses.

Design the subblocks and find out how many addresses are still available after these allocations.

8. (a) Expand the IPv6 address 0:15::1:12:1213 to its original form. 2

(b) Briefly describe the Address Resolution Protocol (ARP) and Reverse Address Resolution Protocol (RARP) operation with appropriate illustration. 5

(c) A host with IP address 192.23.168.27 and physical address C2:4B:5A:10: 2C:1A has a packet to send to another host with IP address 197.29.32.12 and physical address B4:66: F1:5C:83: A3. The two hosts are on the same Ethernet network. Show the ARP request and reply packets encapsulated in Ethernet frames. 4

(d) What is network address translation (NAT)? Explain the need and process with an example. 3